IT4010 Research Project – 2026 Jul

Topic Assessment Form

V2.2

Project ID: J26-ISE-307

1. Topic (12 words max)

AI-Based Intelligent EV Charging Prediction and Decision Support Framework

2. Research group the project belongs to

CoEAI - Centre of Excellence for AI

3. Specialization of the project belongs to

Information System Engineering (ISE)

4. If a continuation of a previous project:

Project ID	No
Year	No

5. Brief description of the research problem including references (300 words max) – references not included in word count.

The rapid adoption of Electric Vehicles (EVs) has significantly increased the demand for charging infrastructure. However, EV users continue to experience long waiting times, charging station congestion, inaccurate charging duration estimations, uncertainty about battery consumption during trips, unexpected charging costs, and difficulty selecting the most suitable charging station. These challenges reduce user satisfaction, create uncertainty during trip planning, and lead to inefficient utilization of charging stations.

Existing EV charging applications mainly provide static information such as charging station locations, charger availability, and navigation support. Most systems display only the current charging status and require users to manually estimate their battery usage, charging cost, and charging requirements before arriving at a charging station. In addition, existing applications rarely integrate intelligent queue prediction, charging duration prediction, battery consumption prediction, charging cost estimation, and charging station recommendation into a single platform.

To address these limitations, this research proposes an AI-Based Intelligent EV Charging Management System that integrates four intelligent functionalities: AI-based charging demand and queue prediction, AI-based charging duration prediction, AI-based battery consumption prediction and charging cost estimation, and context-aware charging station recommendation with route optimization.

By combining machine learning, predictive analytics, and location-based services, the proposed system forecasts future charging demand, predicts personalized charging duration, estimates battery consumption during the journey, calculates the expected charging cost before the charging session, and recommends the most suitable charging station based on predicted queue conditions, charger availability, travel distance, traffic conditions, and user context. The proposed solution enables EV users to make informed charging decisions in advance, reduces waiting time, improves charging station utilization, and enhances the overall EV charging experience.

References

[1] International Energy Agency (IEA), *Global EV Outlook 2025*, Paris: IEA, 2025.

[2] Zhang, Y., Chen, X., & Wang, J., "Machine Learning-Based Electric Vehicle Charging Demand Forecasting: A Review," *Renewable and Sustainable Energy Reviews*, vol. 188, 2024.

[3] Lee, S., Kim, J., & Park, H., "AI-Based Reservation Scheduling for Electric Vehicle Charging Stations," *IEEE Access*, vol. 12, 2024.

[4] Ghosh, A., Roy, S., & Das, P., "Machine Learning Models for EV Charging Duration Prediction," *Energy AI*, vol. 15, 2024.

[5] Deb, S., Tammi, K., Kalita, K., & Mahanta, P., "Review of Recent Trends in Charging Infrastructure Planning for Electric Vehicles," *Wiley Interdisciplinary Reviews: Energy and Environment*, vol. 7, no. 6, 2024.

6. Brief description of Existing Research and Systems (300 words max)

Research studies and commercial EV charging applications have introduced several intelligent solutions to improve the EV charging experience. Artificial Intelligence (AI) and Machine Learning (ML) techniques have been applied for charging demand prediction, queue prediction, charging duration estimation, battery consumption prediction, charging cost estimation, and charging station recommendation. These studies demonstrate that AI can improve charging efficiency and assist users in making better charging decisions.

Existing commercial EV charging applications mainly provide services such as charging station location, charger availability, navigation, charging status, and basic reservation facilities. Recent research has proposed AI-based models for charging demand prediction, queue prediction, charging duration estimation, battery consumption prediction, charging cost estimation, and charging station recommendation. Different machine learning algorithms, predictive analytics, and optimization techniques have been used to improve charging efficiency and support user decision-making.

However, most existing solutions focus on individual functionalities rather than integrating multiple AI-driven services into a single intelligent charging management platform. In addition, very few studies combine AI-based charging demand and queue prediction, charging duration prediction, battery consumption prediction with charging cost estimation, and intelligent charging station recommendation to support end-to-end charging decision-making before users arrive at a charging station.

To address these limitations, this research proposes an AI-Based Intelligent EV Charging Management System that integrates these four intelligent functionalities into a single mobile application. By combining predictive analytics, machine learning, and location-based services, the proposed system aims to reduce waiting time, improve charging station utilization, estimate charging costs in advance, and enhance the overall EV charging experience.

References

[1] International Energy Agency (IEA), *Global EV Outlook 2025*. Paris, France: IEA, 2025.

[2] A. Almaghrebi, F. Aljuheshi, M. Rafaie, K. James, and M. Alahmad, "Data-Driven Charging Demand Prediction at Public Charging Stations Using Supervised Machine Learning," *Energies*, vol. 13, no. 16, Art. no. 4231, Aug. 2020.

[3] N. Sadeghianpourhamami, N. Refa, M. Strobbe, and C. Develder, "Quantitative Analysis of Electric Vehicle Flexibility: A Data-Driven Approach," *International Journal of Electrical Power & Energy Systems*, vol. 95, pp. 451–462, Feb. 2018.

[4] Open Charge Map, "Open Charge Map – Global Public Electric Vehicle Charging Locations," [Online]. Available: https://openchargemap.org. [Accessed: Jul. 17, 2026].

[5] M. Yilmaz and P. T. Krein, "Review of Battery Charger Topologies, Charging Power Levels, and Infrastructure for Plug-In Electric and Hybrid Vehicles," *IEEE Transactions on Power Electronics*, vol. 28, no. 5, pp. 2151–2169, May 2013.

7. Brief description of the solution's nature, including a conceptual diagram (maximum 500 words).

The proposed research introduces an AI-Based Intelligent EV Charging Management System designed to improve the efficiency, reliability, and convenience of electric vehicle charging by integrating predictive analytics, machine learning, and location-based services into a single intelligent mobile platform.

Unlike existing EV charging applications that mainly provide real-time charger availability and navigation support, the proposed system offers four integrated AI-driven functionalities. The first component predicts future charging demand, queue length, waiting time, and congestion level using historical charging records and temporal patterns. The second component predicts personalized EV charging duration based on vehicle characteristics, battery capacity, current battery level, charger type, and historical charging session data.

The third component predicts battery consumption during the user's journey to the selected charging station by analyzing travel distance, estimated travel time, traffic conditions, vehicle characteristics, and historical driving patterns. Based on the predicted arrival battery level and the user's target battery level, the system estimates the required charging energy and charging cost before the charging session begins, allowing users to make cost-aware charging decisions in advance.

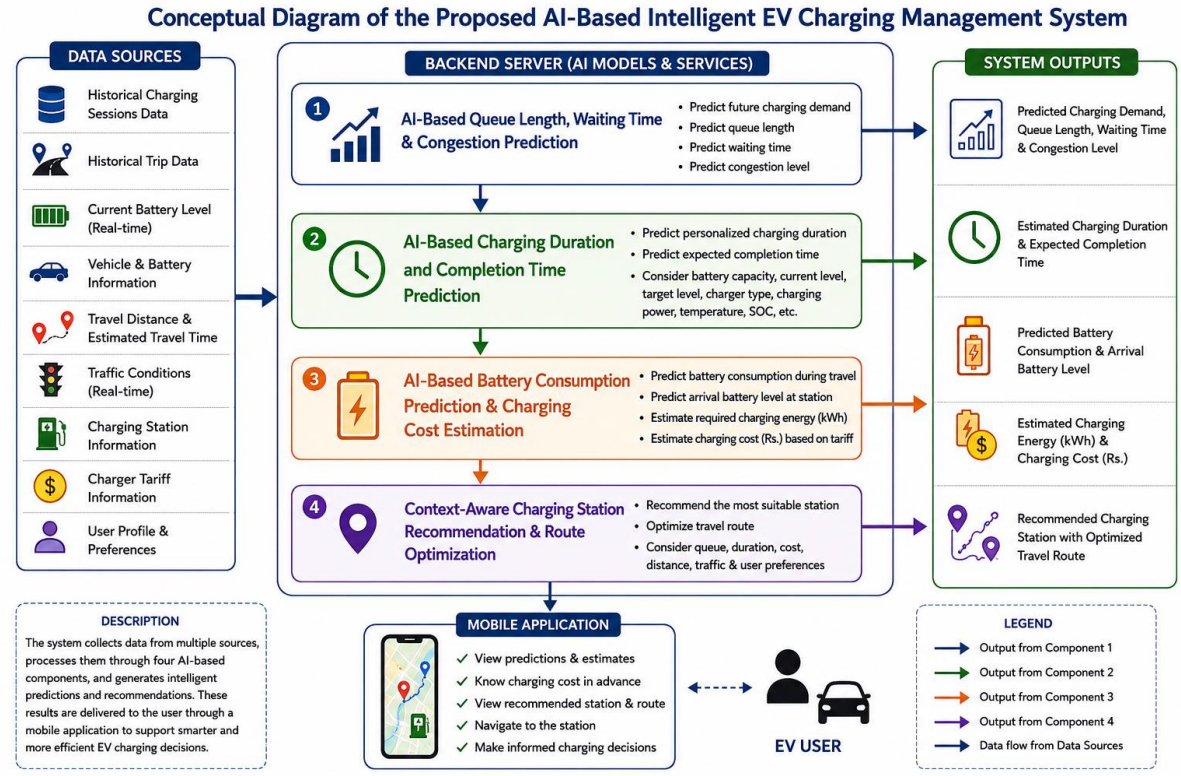
The fourth component recommends the most suitable charging station by considering predicted queue conditions, charger availability, travel distance, traffic conditions, estimated charging cost, and user preferences. This enables users to select the most efficient charging station according to both travel and charging requirements.

The proposed solution provides an end-to-end intelligent decision-support mechanism for EV users before they arrive at a charging station. By integrating multiple AI prediction models with Google Maps services and a centralized backend system, the application minimizes waiting time, improves charging station utilization, predicts charging costs in advance, and enhances the overall EV charging experience.

The system will be implemented as a mobile application connected to a centralized backend server. Machine learning models will process historical charging sessions, trip information, traffic conditions, charging station information, and vehicle-related data to generate intelligent predictions and recommendations. Google Maps API will be integrated to obtain travel distance, estimated travel time, and route information, while a centralized database will store charging sessions, trip history, station details, and prediction results.

System Perspective (Input – Process – Output)

Inputs	Processes	Outputs
Historical charging records	AI-Based Charging Demand and Queue Prediction	Predicted charging demand, queue length, waiting time and congestion level
Current battery level, battery capacity, charger type and vehicle information	Charging Duration Prediction	Estimated charging duration
Current battery level, user location, selected charging station, travel distance, estimated travel time, traffic conditions, vehicle information and historical trip data	AI-Based Battery Consumption Prediction	Predicted battery consumption and predicted arrival battery level
Predicted arrival battery level, target battery level and charger tariff	Charging Cost Estimation	Estimated charging energy (kWh) and estimated charging cost (Rs.)
Charging station information, predicted queue conditions, estimated charging duration, estimated charging cost, traffic conditions and user location	Recommendation Engine and Route Optimization	Personalized charging station recommendation and optimized travel route



8. Brief overview of the availability of necessary specialized domain expertise, knowledge, and data. (500 words max)

The successful implementation of the proposed research requires specialized knowledge in Artificial Intelligence (AI), Machine Learning (ML), predictive analytics, data preprocessing, mobile application development, database management, and Geographic Information Systems (GIS). Technical knowledge of Python, machine learning libraries, Google Maps API integration, and backend technologies will be required to develop, train, and integrate the AI models within the proposed mobile application.

The proposed system requires several types of data to train and evaluate the AI models. These include historical charging session data, charging station information, vehicle specifications, battery capacity, current battery level, historical trip data, travel distance, estimated travel time, traffic conditions, charger tariff information, user charging preferences, and historical charging demand records. Google Maps API will be used to obtain travel distance, estimated travel time, and route information, while publicly available datasets, charging station information, vehicle specifications, and simulated datasets will be used where real-world historical data are unavailable.

The proposed AI models will analyze these datasets to predict charging demand and queue conditions, estimate charging duration, predict battery consumption during travel, estimate charging cost before charging, and recommend the most suitable charging station. Historical charging and trip data will be preprocessed to improve data quality before training and validating the machine learning models.

The research will follow appropriate ethical standards throughout the data collection and system development process. Personal information collected from users will be anonymized to protect user privacy and confidentiality. The proposed system will not disclose personally identifiable information, and all collected data will be used solely for academic research purposes. Appropriate security measures will be implemented to ensure data confidentiality, integrity, and secure storage. In addition, the developed AI models will be evaluated to minimize prediction bias, improve fairness, and provide reliable recommendations for EV users.

9. Objectives and Novelty

Main Objective

To develop an AI-Based Intelligent EV Charging Management System that predicts charging demand, estimates EV charging duration, predicts battery consumption and estimates charging cost, and recommends the most suitable charging station to reduce waiting time, improve charging station utilization, and enhance the overall EV charging experience.

Member	Sub Objective	Tasks	Novelty
S.L.S. Sandeepa	Develop an AI-Based Queue Length, Waiting Time and Congestion Prediction model to accurately predict future queue conditions at EV charging stations.	• Collect and preprocess historical charging station data. • Identify the required data: vehicle arrivals, queue length, available chargers, charging duration, time of day, day type, and holiday information. • Train and evaluate AI models for queue length, waiting time, and congestion prediction. • Generate future queue length, waiting time, and congestion predictions. • Integrate prediction results with the mobile application.	Uses AI to predict future queue length, waiting time, and congestion before users arrive at the charging station by analyzing historical charging patterns, vehicle arrivals, and charger availability. This enables proactive charging decisions instead of relying only on the current charger status.

K.M.S.K. Malshani	Develop an AI-Based Charging Duration Prediction model to accurately predict EV charging duration and estimate the expected charging completion time.	• Identify the factors affecting EV charging duration. • Identify and collect the required data: vehicle model, battery capacity, current battery level, charger type, charging power, and historical charging session data. • Preprocess and analyze charging session data. • Train and evaluate AI models for charging duration prediction. • Predict EV charging duration and estimate the expected charging completion time. • Integrate prediction results with the reservation module.	Uses AI to predict personalized EV charging duration by analyzing vehicle characteristics, battery condition, charger specifications, and historical charging session data, enabling more accurate reservation scheduling than fixed charging time estimates.
G.D.T. Devinda	Develop an AI-Based Battery Consumption Prediction and Charging Cost Estimation model to predict battery consumption during a trip and estimate the charging cost before the charging session.	• Identify the factors affecting EV battery consumption during travel. • Identify and collect the required data: current battery level, vehicle model, battery capacity, travel distance, travel time, traffic conditions, destination charging station, charger tariff, and historical charging session data. • Preprocess and analyze historical trip and charging data. • Train and evaluate AI models for battery consumption prediction.	Uses AI to predict battery consumption and the expected battery level at the selected charging station by analyzing vehicle characteristics, travel distance, traffic conditions, and historical driving patterns. Based on the predicted arrival battery level, the system estimates the required charging energy

		• Predict the expected battery level when the vehicle arrives at the selected charging station. • Estimate the required charging energy and charging cost based on the predicted arrival battery level and the user's target battery level. • Integrate the prediction and charging cost estimation with the mobile application.	and charging cost before the charging session, enabling users to plan charging more efficiently instead of relying on manual estimations.
P.H.T.T. Vijepala	Develop a Context-Aware Charging Station Recommendation and Route Optimization system that recommends the most suitable charging station.	• Collect charging station and traffic information. • Integrate Google Maps API. • Develop a multi-factor recommendation algorithm. • Rank charging stations based on prediction results and user preferences. • Generate optimized travel routes.	Recommends charging stations by combining predicted waiting time, charger availability, traffic conditions, travel distance, and user preferences, rather than recommending only the nearest available station.

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6. **This section is to be completed by the Supervisor and the Co-supervisor of the Project.**

a) This research topic possesses a comprehensive scope suitable for a final-year project.

Yes	✓	No	

b) The proposed topic exhibits novelty.

Yes	✓	No	

c) The student group can successfully execute the proposed project.

Yes	✓	No	

d) The proposed sub-objectives reflect the students' areas of specialization.

Yes	✓	No	

e) Data required for the project is available.

Yes	✓	No	

f) Any other comments:

	Title	First Name	Last Name	Signature
Supervisor*	Mr.	Sanjeewa	Perera	
Co-Supervisor*	Prof	Anuradha	Jayalath	
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				

*- At this stage, identifying the Co-Supervisor is compulsory. Students should reach out to a staff member, with guidance from their supervisor, to serve as the co-supervisor for this project.

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This part is to be filled by the Topic Screening Panel members.

Project ID: J36-IT66-307

Acceptable: Mark/Select as necessary

Topic Assessment Accepted	
Topic Assessment Accepted with minor changes (should be followed up by the supervisor)*	✓
Topic Assessment to be Resubmitted with major changes*	
Topic Assessment Rejected. Topic must be changed	

* Detailed comments given below

Comments

Compond 1 – Novelty is not clearly presented
Data requirements not identified
Componand 2 – Data required for the prediction not identify
Com ponet 3– change (Member absend)
Componerd 4 – good.

The Review Panel Details

Member's Name	Signature	Date
Chathurangika Kahandawaarachchi		27/07/2026.

***Important:**

1. According to the comments given by the panel, make the necessary modifications and get approval from the **Supervisor** or the **Same Panel.**
2. If the project topic is rejected, identify a new topic, and follow the same procedure until the topic is approved by the assessment panel.